

CHAPTER II-1

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2. (a) Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves on X , and $p \in X$ a point. Show that $(\ker \varphi)_p \cong \ker(\varphi_p)$ and $(\operatorname{im} \varphi)_p \cong \operatorname{im}(\varphi_p)$.
 (b) Show that φ is injective (resp. surjective) if and only if φ_p is injective (resp. surjective) for all $p \in X$.
 (c) Show that a sequence $\cdots \rightarrow \mathcal{F}_{i-1} \xrightarrow{\varphi_{i-1}} \mathcal{F}_i \xrightarrow{\varphi_i} \mathcal{F}_{i+1} \rightarrow \cdots$ of sheaves is exact if and only if, for all $p \in X$, the associated sequence of stalks is exact.

Proof. (a) A germ $(s, U) \in (\ker \varphi)_p \subset \mathcal{F}_p$ iff there exists $p \in V \subset U$ with $\varphi(U)(s)|_V = 0$. That is,

$$\varphi_p(s, U) = (\varphi(V)(s)|_V, V) = (0, V).$$

Similarly, a germ $(s, U) \in (\operatorname{im} \varphi)_p \subset \mathcal{G}_p$ iff there exists $p \in V \subset U$ with $s|_V = \varphi(V)(t)$. That is,

$$(s, U) = (s|_V, V) = (\varphi(V)(t), V) = \varphi_p(t, V).$$

(b) φ is injective iff $\ker \varphi \cong 0$. The zero map $\ker \varphi \rightarrow 0$ is an isomorphism iff each of its localizations $(\ker \varphi)_p \rightarrow 0$ are isomorphisms. Using (a), we see that φ is injective iff $\ker(\varphi_p) \cong (\ker \varphi)_p \cong 0$, i.e. φ_p injective, for all $p \in X$.

φ is surjective iff $\mathcal{G} = \operatorname{im} \varphi$. That is, the inclusion map $i : \operatorname{im} \varphi \rightarrow \mathcal{G}$ is an isomorphism. This occurs iff each $(\operatorname{im} \varphi)_p \rightarrow \mathcal{G}_p$ is an isomorphism, which occurs iff each $\operatorname{im}(\varphi_p) \xrightarrow{\sim} (\operatorname{im} \varphi)_p \rightarrow \mathcal{G}_p$ is an isomorphism. This map is the inclusion $\operatorname{im}(\varphi_p) \rightarrow \mathcal{G}_p$, so the result follows.

(c) It suffices to show the result for short exact sequences $0 \rightarrow \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \rightarrow 0$. (b) gives us φ injective iff each φ_p is injective, ψ surjective iff each ψ_p is surjective.

If $\operatorname{im} \varphi = \ker \psi$, then this induces natural isomorphisms $\operatorname{im} \varphi_p \cong \ker \psi_p$ for all $p \in X$. Inclusions of sheaves induce inclusions of stalks, so $\operatorname{im} \varphi_p = \ker \psi_p$.

Suppose instead that $\operatorname{im} \varphi_p = \ker \psi_p$ for all $p \in X$, and fix $U \subset X$, $s \in \mathcal{F}'(U)$. For each $p \in U$, we have $\widetilde{\operatorname{im} \varphi_p} = \ker \psi_p$, so $\psi_p \varphi_p(s, U) = (0, U)$. That is, there is $p \subset V_p \subset U$ such that

$$\psi(U) \varphi(U)(s)|_{V_p} = 0.$$

Then the V_p form a cover of U , so the sheaf axiom gives us $\psi(U) \varphi(U)(s) = 0$, whence

$$\widetilde{\operatorname{im} \varphi} \subset \ker \psi.$$

Then, in fact, there is an inclusion $i : \operatorname{im} \varphi \hookrightarrow \ker \psi$ which induces the inclusions $i_p : \operatorname{im} \varphi_p \hookrightarrow \ker \psi_p$ for each $p \in V$. The latter inclusions are isomorphisms, so it follows that the former is an isomorphism as well. That is,

$$\operatorname{im} \varphi = \ker \psi.$$

□

8. Show that, for any $U \subset X$ open, the functor $\Gamma(U, -) : \operatorname{Sh}(X) \rightarrow \mathfrak{Ab}$ is left-exact.

Proof. Suppose $0 \rightarrow \mathcal{F}' \xrightarrow{\varphi} \mathcal{F} \xrightarrow{\psi} \mathcal{F}'' \rightarrow 0$ is exact. Then $\ker(\varphi(U)) = (\ker \varphi)(U) = 0$ gives us that $\varphi(U)$ is injective. It suffices to show that $\widetilde{\text{im}(\varphi(U))} = \ker(\psi(U))$. Since $\ker \psi = \text{im } \varphi = (\widetilde{\text{im } \varphi})^+$, it suffices to show that $\widetilde{\text{im } \varphi}$ is a sheaf.

Let $\{V_i\}$ cover $V \subset X$ open, and $s_i \in \text{im}(\varphi(V_i))$ be sections with $s_i|_{V_{ij}} = s_j|_{V_{ij}}$ for all i, j . $\varphi(V_i)$ injective gives unique $t_i \in \mathcal{F}'(V_i)$ such that $s_i = \varphi(V_i)(t_i)$, and

$$\varphi(V_{ij})(t_i|_{V_{ij}}) = s_i|_{V_{ij}} = s_j|_{V_{ij}} = \varphi(V_{ij})(t_j|_{V_{ij}})$$

implies that the restrictions of our t_i, t_j agree as well. Then $\mathcal{F}'$ a sheaf gives us a unique $t \in \mathcal{F}'(V)$ with $t|_{V_i} = t_i$. Let $s = \varphi(V)(t) \in \widetilde{\text{im } \varphi}(V)$ be its image, so that $s|_{V_i} = \varphi(V_i)(t|_{V_i}) = s_i$ for each i . This shows existence.

Suppose we have another $s' = \varphi(V)(t')$ with $s'|_{V_i} = s_i$ for each i . Then injectivity and $\varphi(V_i)(t'|_{V_i}) = \varphi(V_i)(t|_{V_i})$ implies $t'|_{V_i} = t|_{V_i}$ for all i , whence $t' = t$, and $s' = s$. We conclude that s is unique, so that $\widetilde{\text{im } \varphi}$ is a sheaf. $\square$